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GUTKOWSKI et al.(10) **Pub. No.: US 2025/0277372 A1**(43) **Pub. Date: Sep. 4, 2025**(54) **SET OF FLOOR PANELS****Publication Classification**(76) Inventors: **Pawel GUTKOWSKI**, Barlinek (PL);
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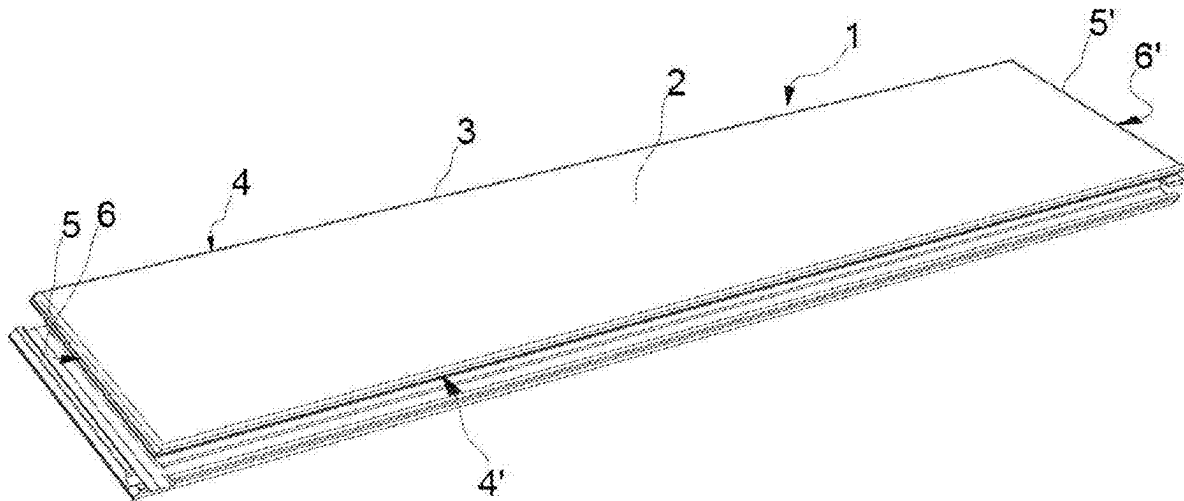
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(57) **ABSTRACT**

The floor panel (1) comprises a concave locking part (11) with an upper retaining protrusion (21) and with a lower retaining protrusion (22), in a male joining part (9), and a convex locking part (12) with an upper retaining surface (17) and with a lower retaining surface (24), in a female joining part (10). Furthermore, the male joining part (9) of the floor panel (1) has a sealing edge (16) shaped at its outer side surface (6'), wherein, in the locked state, this sealing edge (16) of the male joining part (9), when engaged with the female joining part (10) of another floor panel (1), is pressed down with a fixed force into the upper retaining surface (17) of the convex locking part (12).



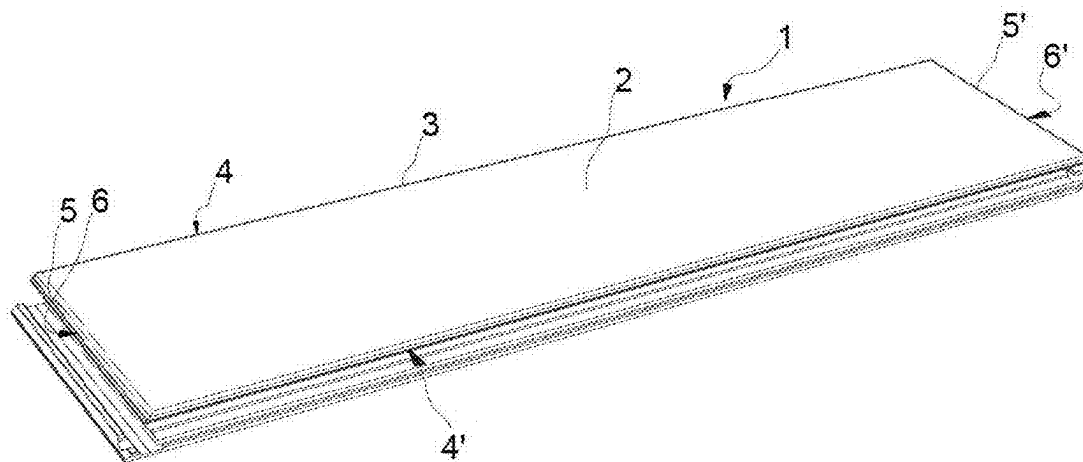


Fig. 1

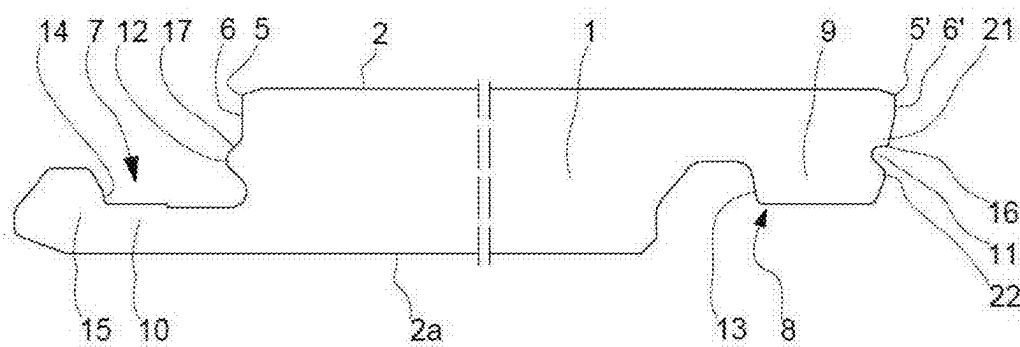


Fig. 2

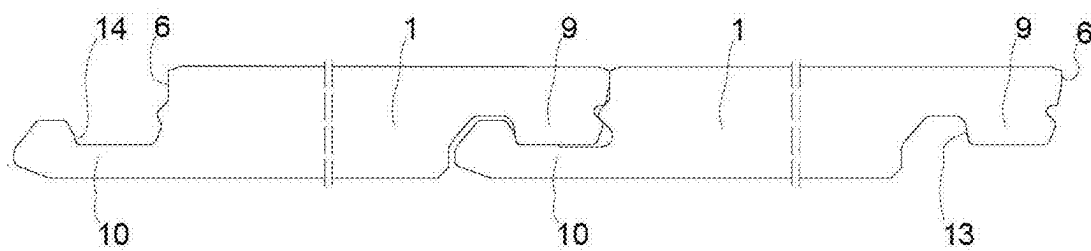


Fig. 3

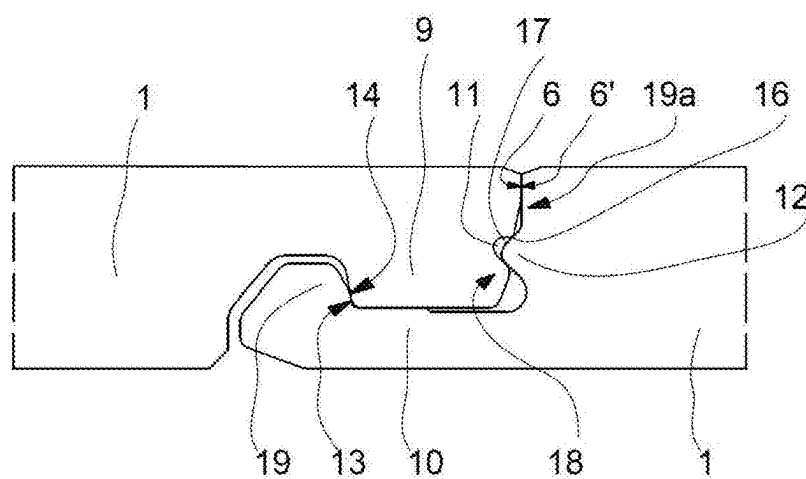
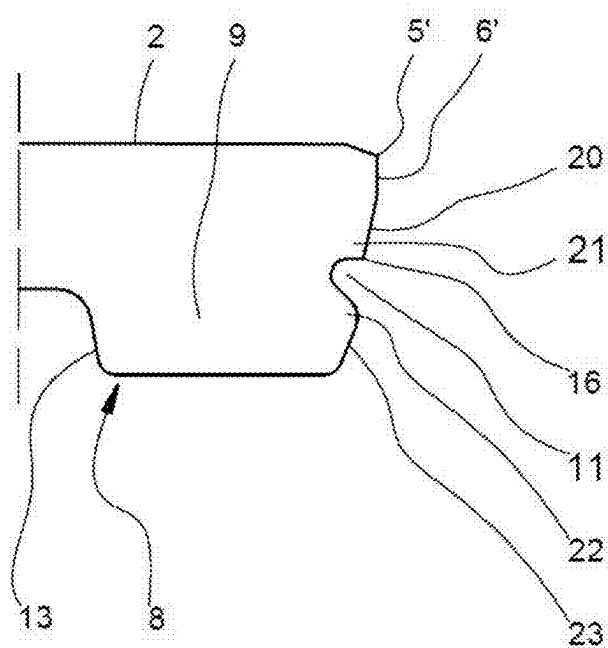


Fig. 4



SET OF FLOOR PANELS

[0001] This invention relates to a set of floor panels, designed for watertight joining of joints of adjacent panels. The invention is applicable to floor panels made of solid wood, engineered-wood floor panels, laminated wood-based panels and floor panels made of plastic conglomerates.

[0002] Floor panels, in particular those made of wood-based materials, are not resistant to changes in humidity and temperature. Changes in microclimate cause the wood to swell or dry out, which causes its dimensions to change. The percentage change in dimensions of individual wood layers of a floor panel is highly dependent on its fibre orientation. The wood of species commonly used in production of floor panels, when the fibre orientation is longitudinal, changes its dimensions only slightly when exposed to moisture, whereas such with a radial or tangential fibre orientation significant changes may occur in the dimensions of the wood due to increased moisture. Furthermore, the compressive strength of the wood in the radial or tangential direction is approximately five times lower compared to the compressive strength of wood with the longitudinal fibre orientation. The high compressive strength in the longitudinal direction is related to increased brittleness of wood, which results in localised, spalling damage to wood fibres in floor panels, near the edges of the short sides when the wood is swollen. On the other hand, in the case of the radial or tangential fibre orientation, low pressure resistance combined with ductility can be the cause of plastic deformations known as edging. To prevent damage to the panels due to swelling of the wood when exposed to moisture, suitable expansion gaps are designed at the joints of the adjoining edges of the upper layers.

PRIOR ART

[0003] Moisture penetration from the upper surface of the panel through expansion gaps and joints towards the lower surface of the panel is an unfavourable phenomenon, since the moisture accumulated in the lower parts of the floor panel can cause the absorbent layers to swell and thus cause unfavourable stresses and the resulting bulging of the panels. Even in the case of panels made of waterproof or non-absorbent plastic material, the moisture penetration through joints can also cause the occurrence of unfavourable phenomena, such as growth of bacteria, fungi or mould, located in inaccessible areas of the joints or under the laid floor.

[0004] The known snap-fit joining of floor panel sets disclosed in WO0047841 includes a vertical fixing assembly for locking floor panels being joined in a direction perpendicular to the upper surface of the panel and has joining parts made as a male joining part and a female joining part, which are equipped vertically with an active concave locking part and a convex locking part. In addition to fulfilling the basic function of a snap-fit joining, these locking parts have slanted surfaces elastically pressed against each other, which, in addition to their locking function, can also fulfil the function of a sealing assembly designed to prevent moisture from the upper part of the panel penetrating into its subfloor.

[0005] From WO03016654 there is known a joint for rectangular floor panels comprising engaging elements parallel to the upper surface and protruding horizontally beyond that upper surface. The longitudinal engaging elements

further have receiving recesses for engaging with protrusions of adjacent floor panels. The engaged panels form together a horizontal fixing assembly and a vertical fixing assembly. Furthermore, the longitudinal engaging element has, at its end, a retaining surface spaced away from the retaining surface of the adjacent floor panel, thereby creating an expansion gap. In particular, the known floor panel joint, in an alternative embodiment, has a vertical fixing assembly with a locking assembly using an additional elastic locking element with a circular cross-section, which at the same time can fulfil the function of preventing moisture from entering the lower zones of the joined panels.

[0006] In EP1971736B1, the joints of the floor panels are sealed with a sealant. Sealing means may also include a horizontally running protrusion that engages in an open side groove. This sealing system in the form of a horizontal protrusion cooperating with a horizontal groove may additionally contain a sealant contained in the horizontal groove. The protrusion and the groove can also be arranged vertically and constitute a hook sealing joint in this form.

[0007] Also known from EP1412596B1 is a floor panel with a seal protecting against moisture penetration at the border of joined panels. To ensure that the lower layers of the joined panels are protected, the protruding part of the joint of one panel is pressed against the shape-matched opposite joined surfaces of the adjacent panel.

DISCLOSURE OF THE INVENTION

[0008] The object of the invention is to develop a set of floor panels in which the design of latching means ensures their easy joining, vertical and horizontal fixing, as well as improves the tightness and compactness of the joint.

[0009] In the set of floor panels according to the invention the joined floor panels are in the form of rectangular plates and comprise

- [0010] an upper surface and a lower surface parallel to it, designed to be laid on a subfloor,
- [0011] two long edges of side surfaces,
- [0012] two short edges of side surfaces,
- [0013] complementary latching means shaped near the edges being joined to each other,
- [0014] a vertical fixing assembly,
- [0015] a horizontal fixing assembly.

[0016] The vertical fixing assembly serves to lock the floor panels being joined in a direction perpendicular to the upper surface of the panel and comprises joining parts made as a male joining part and a female joining part, which have vertically an active concave locking part and a convex locking part. The concave locking part has an upper retaining protrusion and a lower retaining protrusion, and the convex locking part has an upper retaining surface and a lower retaining surface.

[0017] In addition, the upper retaining protrusion of the concave locking part in the locked state rests on the upper retaining surface of the convex locking part, and the lower retaining protrusion of the concave locking part rests on the lower retaining surface of the convex locking part.

[0018] The horizontal fixing assembly serves to lock the floor panels being joined in a direction parallel to the upper surface of the panel and comprises a horizontally active inner side surface of the male joining part, wherein this inner side surface, in the locked state, rests against an inner retaining surface of a protruding engaging element and comprises a horizontally active outer side surface of the

male joining part, wherein this outer side surface, in the locked state, rests against an inner retaining surface of the female joining part.

[0019] The solution according to the invention is characterised in that the male joining part of the floor panel has a sealing edge shaped at its outer side surface, wherein in the locked state this sealing edge of the male joining part, when engaged with the female joining part of another floor panel, is pressed down with a fixed force into the upper retaining surface of the convex locking part.

[0020] Preferably, the sealing edge is shaped on the upper retaining protrusion of the concave locking part.

[0021] Preferably, the sealing edge is shaped by the inclined upper outer surface of the guiding male joining part.

[0022] Preferably, the sealing edge has a rounding radius of less than 0.3 mm.

[0023] Preferably, the sealing edge has a rounding radius within 0.1 to 0.2 mm.

[0024] The invention solves the problems associated with moisture penetration from the upper side of the floor panel through the joint assembly.

[0025] The shaping of the sealing edge on the outer side of the male joining part of the floor panel allows for small force to be exerted to achieve tightness of the joining parts. With fairly low pushing forces, high pressure acting on the surfaces is achieved, resulting in better sealing against moisture penetration into the lower regions of the joint and under the lower surface of the panel.

[0026] The pressing of the sealing edge into the upper retaining surface of the convex locking part with a predetermined force is intended to achieve a tight joint even in case of certain imperfections in the linearity of said sealing edge or in case of imperfections in the shape of the upper retaining surface of the convex locking part.

BRIEF DESCRIPTION OF DRAWINGS

[0027] The subject of the invention is shown in its embodiments in drawings, in which:

[0028] FIG. 1 shows the floor panel in a top view,

[0029] FIG. 2 shows the floor panel in cross section,

[0030] FIG. 3 shows a set of the floor panels in the locked state,

[0031] FIG. 4 depicts an enlarged part of the floor panel set from FIG. 3, showing complementary latching means in the locked state,

[0032] FIG. 5 depicts an enlarged detail showing the latching means in the form of a male joining part,

[0033] FIG. 6 depicts an enlarged detail showing the latching means in the form of a female joining part,

[0034] FIG. 7 shows an enlarged detail of the joint between the concave locking part and the convex locking part in the locked state.

BEST MODE FOR CARRYING OUT THE INVENTION

[0035] A floor panel 1 shown in FIG. 1 is in the form of a rectangular plate with a core made of polyvinyl chloride and chalk conglomerate. The upper surface 2 of the floor panel 1 is bounded by two long edges 3, 3' of side surfaces 4, 4' and two short edges 5, 5' of side surfaces 6, 6'.

[0036] As shown in FIG. 2, the floor panel has complementary latching means 7, 8 shaped near the short edges 5, 5' joining each other during assembly. These latching means

7, 8, after joining a male joining part 9 with a female joining part 10, lock the floor panels 1 being joined in a direction perpendicular to the upper surface 2 and in a direction parallel to either this upper surface or to a lower surface 2a. For locking in the perpendicular direction, the male joining part 9 has a vertically active concave locking part 11 and the female joining part 10 has a vertically active convex locking part 12.

[0037] In order to lock the floor panels 1 being joined in a direction parallel to the upper surface 2, the male joining part 9 has a horizontally active inner side surface 13 and the horizontally active outer side surface 6', and the female joining part 10 has an inner retaining surface 14 and the horizontally active side surface 6, with the inner retaining surface 14 being shaped on a protruding engaging element 15. Furthermore, a sealing edge 16 is formed on the male joining part 9, on the outer side surface 6', at the boundary with the concave locking part 11, intended to be pressed into an upper retaining surface 17 of the convex locking part 12.

[0038] FIG. 3 and FIG. 4 show a set of two identical floor panels 1 in the locked state. In such a state, the male joining part 9 is coupled to the female joining part 10 and the convex locking part 12 is inserted into the concave locking part 11. The stable locking of the joined panels 1 is realised by a vertical fixing assembly 18, in a direction perpendicular to the upper surface 2, by the concave locking part 11 and the convex locking part 12, and by a horizontal fixing assembly 19 and 19a in a direction parallel to the upper surface 2, wherein the said horizontal fixing assembly 19, 19a comprises the inner side surface 13 of the male joining part 9 being in contact with the inner retaining surface 14 of the female joining part 10, and further comprises the outer side surface 6' of the male joining part 9 being in contact with the side surface 6 of the female joining part 10. With such a snap-fit joining of the floor panels 1 there is a linear thrust of the sealing edge 16 against the upper retaining surface 17 of the convex locking part 12. The small contact area between the sealing edge 16 and the upper retaining surface 17 results in a presence of high pressure at this contact area and thus ensures increased joint tightness.

[0039] As shown in FIG. 5 and FIG. 6, the latching means 8 is shaped like a downward facing projection in the form of the male joining part 9 and is dimensionally adapted to the latching means 7 in order to be snap-fitted to the female joining part 10.

[0040] As shown in FIG. 5, the male joining part 9 has a flat outer side surface 6' transitioning into an inclined upper outer guiding surface 20 forming the sealing edge 16 in an upper retaining protrusion 21, at the joint with the upper part of the concave locking part 11. The lower part of the concave locking part in the form of a lower retaining protrusion 22 transitions archwise into an inclined lower outer guiding surface 23 which transitions gently into the lower surface 2a of the male joining part 9.

[0041] In the presented embodiment, where the core of the panel is made of polyvinyl chloride and chalk conglomerate, the sealing edge 16 preferably has a rounding radius within 0.15 mm with a tolerance of ± 0.02 mm. This rounding radius of the sealing edge 16 can also be used with a positive effect for floor panels made of solid wood, engineered wood floor panels, laminated wood-based panels and floor panels made of other plastic conglomerates. Considering the possibility

of increased tightness of the floor panel joints, the dimensional limit of the rounding radius of the sealing edge 16 should not exceed 0.3 mm.

[0042] The rounding of the sealing edge 16 slightly increases the contact area with the upper retaining surface 17 of the convex locking part 12, and thus reduces the pressure of this slightly rounded sealing edge 16 in contact with the upper retaining surface 17. It is also possible to use a sharp sealing edge 16 without rounding it, in the shape formed directly by milling the concave locking part 11 and the adjacent upper outer guiding surface 20. However, even a slight rounding of the sealing edge largely increases the resistance to its possible damage that may occur during the floor laying stage.

[0043] As shown in FIG. 6, the female joining part 10 has the flat inner side surface 6 transitioning into the convex locking part 12 having the upper retaining surface 17 and a lower retaining surface 24. The lower retaining surface 24 transitions through an arc 25 into an undercut 26 of a lower seating surface 27 which, in turn, transitions through an arc 28 into the inner retaining surface 14 and further into an inner guiding surface 29.

[0044] An enlarged detail of the joint between the concave locking part 11 and the convex locking part 12 in the locked state is shown in FIG. 7. In this state, when the male joining part 9 is joined to the female joining part, the convex locking part 12 enters the concave locking part 11. With such a coupling, the upper retaining protrusion 21 of the concave locking part 11 rests with its sealing edge 16 on the upper retaining surface 17 of the convex locking part 12, and the lower retaining protrusion 22 of the concave locking part 11 rests on the lower retaining surface 24 of this convex locking part 12. By selecting the dimensional tolerance of the concave locking part 11 between the contact points with the convex locking part 12, taking into account the elasticity of the core material of the floor panel 1, it is possible to determine the pressure force acting at these contact points. In the locked state, the sealing edge 16 also exerts this force on the upper retaining surface 17 of the convex locking part 12, thereby enhancing the tightness of the set of joined floor panels 2.

LIST OF REFERENCE NUMBERS

[0045]	1 Floor panel;
[0046]	2 Upper surface;
[0047]	2a Lower surface;
[0048]	3, 3' Long edges;
[0049]	4, 4' Side surfaces of the long edges;
[0050]	5, 5' Short edges;
[0051]	6, 6' Side surfaces of the short edges;
[0052]	7, 8 Latching means;
[0053]	9 Male joining part;
[0054]	10 Female joining part;
[0055]	11 Concave locking part;
[0056]	12 Convex locking part;
[0057]	13 Inner side surface;
[0058]	14 Inner retaining surface;
[0059]	15 Engaging element;
[0060]	16 Sealing edge;
[0061]	17 Upper retaining surface;
[0062]	18 Vertical fixing assembly;
[0063]	19, 19a Horizontal fixing assembly;
[0064]	20 Upper outer guiding surface;
[0065]	21 Upper retaining protrusion;

[0066]	22 Lower retaining protrusion;
[0067]	23 Lower outer guiding surface;
[0068]	24 Lower retaining surface;
[0069]	5 25 Arc;
[0070]	26 Undercut;
[0071]	27 Lower seating surface;
[0072]	28 Arc;
[0073]	29 Inner guiding surface.

1. A set of floor panels, wherein the joined floor panels (1) are in the form of rectangular plates comprising:

- an upper surface (2) and a lower surface (2a) parallel to it, designed to be laid on the subfloor,
- two long edges (3, 3') of side surfaces (4, 4')
- two short edges (5, 5') of side surfaces (6, 6')
- complementary latching means (7, 8) shaped near the edges (3, 3'; 5, 5') being joined to each other,
- vertical fixing assembly (18),
- horizontal fixing assembly (19, 19a),

wherein the vertical fixing assembly (18), serving to lock the floor panels (1) being joined in a direction perpendicular to the upper surface (2) of the panel (1), comprises joining parts made as a male joining part (9) and a female joining part (10), which have a vertically active concave locking part (11) and a convex locking part (12),

where the concave locking part (11) has an upper retaining protrusion (21) and a lower retaining protrusion (22),

and the convex locking part (12) has an upper retaining surface (17) and a lower retaining surface (24),

wherein the upper retaining protrusion (21) of the concave locking part (11) in the locked state rests on the upper retaining surface (17) of the convex locking part (12), and the lower retaining protrusion (22) of the concave locking part (11) rests on the lower retaining surface (24) of the convex locking part (12),

and the horizontal fixing assembly (19, 19a) serving to lock the floor panels (1) being joined in a direction parallel to the upper surface (2) of the panel (1) comprises a horizontally active inner side surface (13) of the male joining part (9), wherein this inner side surface (13), in the locked state, rests against an inner retaining surface (14) of a protruding engaging element (15) and comprises a horizontally active outer side surface (6') of the male joining part (9), wherein this outer side surface (6'), in the locked state, rests against an inner side surface (6) of the female joining part (10), characterised in that the male joining part (9) of the floor panel (1) has a sealing edge (16) shaped at its outer side surface (6'),

wherein, in the locked state, this sealing edge (16) of the male joining part (9), when engaged with the female joining part (10) of another floor panel (1), is pressed down into the upper retaining surface (17) of the convex locking part (12) with a fixed force which achieves a tight joint, even in case of certain imperfections in the linearity of the sealing edge (16) or in the shape of the upper retaining surface (17) of the convex locking part (12),

wherein the sealing edge (16) is shaped by the inclined upper outer guiding surface (20) of the male joining part (9).

2. The set of floor panels according to claim 1, characterised in that the sealing edge (16) is shaped on the upper retaining protrusion (21) of the concave locking part (11).

3. (canceled)

4. The set of floor panels according to claim 1, characterised in that the sealing edge (16) has a rounding radius of less than 0.3 mm.

5. The set of floor panels according to claim 4, characterised in that the sealing edge (16) has a rounding radius within the range from 0.1 to 0.2 mm.

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